

Algorithmic and High-Frequency Trading

Learning Objectives:

- Algorithmic trading and high-frequency trading
- Types of algorithmic trading strategies
- Effects of algorithmic trading on liquidity and price discovery
- Is there too much investment in trading speed?
- Algorithmic trading and financial markets stability
- Market design and trading speed

9.1 Introduction

As described in Section 1.4.3, trading in financial markets has become increasingly automated and often occurs at phenomenal speed, as a result of progress in information and computing technologies. This evolution raises questions. Many have voiced concerns about possible negative effects of high-frequency trading on market liquidity, price efficiency and informativeness, as well as financial market stability. For instance, the best-selling writer Michael Lewis argues that U.S. equity markets are “rigged” in favor of high-frequency traders (HFTs) at the expense of slower investors in his popular book titled *Flash Boys: A Wall Street Revolt*. Others debate the role that automation and HFTs may have played in several market disruptions such as the flash crash in U.S. stock markets of May 6, 2010, in U.S. treasuries on October 15, 2014, and in the British pound market on October 7, 2016.

These concerns have led to proposals to slow down markets. For instance, some trading platforms (e.g., IEX in the U.S. or Alpha in Canada) now use “speed bumps” that delay the time elapsed between the arrival of an order at a trading platform and its entry into the trading platform’s matching engine. Alternatively, Budish, Cramton, and Shim (2015) propose a switch from continuous trading (the current norm for electronic markets) to frequent batch auctions, in which market clearing would take place at discrete points in time (say, every few seconds). Yet another proposal is to introduce a tax on security transactions (known as a Tobin tax), falling disproportionately on HFTs, due to

their extremely high trading frequency. Such a tax has already been implemented in France in 2012 and in Italy in 2013.

To understand these debates, we start in Section 9.2 by describing the main trading strategies used by high-frequency traders. Then, in Section 9.3, we use the concepts presented in previous chapters to analyze the channels through which high-frequency trading could affect liquidity and price informativeness, and present the main empirical findings about the effects of algorithmic and high-frequency trading. In Section 9.4, we study speculators' decision to invest in speed-enhancing technologies and the effects of such investment on liquidity. Section 9.5 inquires whether computerized trading is destabilizing, in the sense that it amplifies price movements, increases market vulnerability to manipulation and/or transmits mispricing more rapidly across markets. In light of these models, Section 9.6 discusses the main policy interventions that have been proposed to alleviate the potentially harmful effects of HFTs.

9.2 Trading Strategies

Algorithmic trading relies on sophisticated computer programs to generate, route, and execute orders so as to implement a variety of trading strategies.¹ Importantly, not all algorithmic trading occurs at high frequencies. Only a subset of algorithmic traders use strategies that require very fast reaction (in the order of a few milliseconds) to market events (such as quote updates or public news arrivals) and are called high-frequency traders for this reason.² Algorithmic traders' strategies can be classified into four broad types (SEC 2010), the first three of which are widely used by HFTs:

1. *Passive market making*: the submission of non-marketable buy and sell limit orders, providing liquidity to market participants, very much like traditional dealers do. However, high-frequency market makers use highly automated routines for the submission and revision of their quotes (based on inventory exposure, price signals from other markets, etc.), and limit their activity to electronic trading platforms.

¹ This definition encompasses very different types of activities, such as quantitative asset management, agency trading (the placement of orders on behalf of other traders) or high-frequency trading.

² There is no statutory or regulatory definition of high-frequency trading. Well-known high-frequency trading firms include Virtue Financial, Citadel Securities, Two Sigma Securities, Jump Trading, FlowTraders, or Optiver. Some are standalone for-profit companies (like Virtue Financials) while others (like Citadel Securities or Two Sigma Securities) are part of broader financial institutions with asset management activities.

2. *Arbitrage*: exploiting price discrepancies between related securities. Algorithmic trading can enable one to detect and benefit from discrepancies between the prices of related securities (e.g., derivatives and their underlying securities or exchange rates that should be tied by triangular arbitrage) before other market participants. An obvious example of arbitrage opportunities is the case of a stock traded on two platforms, say the French stock Sanofi, which trades on both Cboe Europe and Euronext. If the bid price for Sanofi on Cboe Europe exceeds the ask price on Euronext, it is profitable to buy it on Euronext and sell it on Cboe Europe.³ Speed is of the essence, though, as the profit will be locked in only if the two transactions are virtually simultaneous. Such straightforward arbitrage opportunities are rare and fleeting, precisely because participants take advantage of them in a split second: for instance, Chaboud, Chiquoine, Hjalmarsson, and Vega (2014) find that most arbitrage opportunities on the foreign exchange market last for less than a second.
3. *Directional trading*: aiming to exploit information not yet reflected in prices. The informational advantage of investors can be very fleeting. For instance, consider a news release that leads investors to mark up the value of a firm. Investors who can place buy market orders almost instantaneously can profit by picking off “stale” sell limit orders whose prices do not yet reflect the new value. Again, speed is of the essence because investors who trade on such information can make a profit only by moving before traders on the other side cancel their limit orders or a competing buyer materializes. This “rat race” between directional HFTs is a salient issue in the public debate about HFTs and is analyzed in detail in Section 9.4. Nowadays directional algorithms can exploit an array of techniques to search, extract, and process information: for instance, they can scan the Internet for relevant keywords, and use machine learning techniques to analyze big data or apply textual analysis to the statements of top managers or firms’ regulatory filings (e.g., 10-K or 8-K forms in the U.S.).⁴
4. *Execution algorithms*: implementing trading strategies that reduce the execution cost for large orders. Buy-side institutional trading desks use in-house algorithms or suites of algorithmic execution services provided

³ Wah (2016) shows that arbitrage opportunities of this type are frequent for a large sample (495) U.S. stocks. For these stocks, she finds an average of 69 arbitrage opportunities per day. These opportunities are very short-lived (their median duration is 870 milliseconds). The average potential profit per opportunity is small (\$81) but as arbitrage opportunities are frequent, they generate a large aggregate profit of \$3 billion.

⁴ For instance, Barbopoulos et al. (2021) find that a large number of downloads of new regulatory filings on the SEC Edgar website just after they appear on this site stem from machines. This number has increased over time from less than 1 million in 2008 to more than 8 million in 2016.

by brokerage firms when they rebalance their portfolios. This rebalancing often involves the submission of large orders (“parent orders”) that algorithms optimally split in smaller orders (“child orders”) both over time and across trading platforms in order to minimize portfolio managers’ trading costs. Brokers often provide off-the-shelf algorithms such as “Volume-weighted average price” (VWAP) (which split orders to target the VWAP over a certain period of time; see Chapter 2), “Minimum Implementation Shortfall” (which split orders to minimize the implementation shortfall on a given parent order; see Chapter 2), or “Percent of Volume” (which split orders to target a certain fraction of participation to total trading volume per unit of time).⁵ These algorithms often use limit orders (see Beason and Wahal 2019) and continuously adjust the order placement strategy given changing market conditions (as formalized by Bertsimas and Lo 1998 or Huberman and Stanzl 2007). They also use Smart Order Routing Systems to optimally navigate in fragmented markets, by identifying quickly where are the best possible prices for executing their orders.

This evolution has led to important structural changes in the provision of trading services and financial information. Firstly, algorithmic trading is possible only to the extent that trading platforms are electronic. Thus, algorithmic trading has been enabled by the gradual electronification of trading (that started in the 1960s in equities markets) and in turn has accelerated this evolution in instruments that used to trade over the counter (e.g., currencies and more recently Treasury and corporate bonds).

Moreover, algorithmic trading has triggered a considerable increase in the volume of orders relative to trading volume on trading platforms, forcing the latter to improve the capacity of their matching engines to process very high traffic. In particular, algorithmic trading is often associated with the placement of a large number of limit orders that are subsequently cancelled and repriced. As a result the cancel-to-trade ratio (the number of cancellation divided by the number of trades) has considerably increased over time. This ratio can vary enormously. For instance, between 2013 and 2020 the average daily value of this ratio ranged from 9 (in 2020) to more than 33 (in 2015) for U.S. stocks.⁶

Lastly, the profitability of high-frequency traders’ trading strategies critically depends on their speed of access to information about market events (a quote

⁵ Institutional investors often allocate trades to brokers using an “algo wheel,” which enables them to track the performance of the algorithms provided by different brokers and pick these algorithms optimally depending on the characteristics of their trades and specific constraints. See Johnson (2019), “Trends in global equity execution,” Greenwich Associates.

⁶ See <https://www.sec.gov/marketstructure/> to view various statistics of interest for U.S. stocks and trading platforms.

update, a new trade, a corporate announcement about a stock, etc.) and their ability to minimize communication times (“latencies”) with trading platforms. To do this, they rely on in-house algorithms and computing power but also on external technology providers. First, they buy co-location services from trading platforms, that is, the ability to locate their computers right next to trading platforms’ matching engines and connect these computers by subscribing to high-speed communication services. These services are provided by specialized companies that build infrastructure catering specifically to high-frequency traders. This has led to massive investments. For instance, the Spread Networks fiber-optic cable between the Chicago area (where many U.S. futures markets are located) and New Jersey (where stock and ETF markets are located in the U.S.) alone required a \$300 million investment, and setting up 15 different microwave networks linking the same areas costed about \$120 million. There are also plans to use low earth orbit satellites to transmit information faster between Wall Street and the City of London, requiring investments of at least \$3 billion (see Shkilko and Sokolov 2020). This raises questions about the social benefits of these investments and whether they are not excessive—a point analyzed in detail in Section 9.4.

9.3 Algorithmic Trading and Market Quality

9.3.1 Liquidity

Does algorithmic trading, and more specifically high-frequency trading, affect market liquidity and, if so, why? Addressing this question requires understanding how algorithmic trading affects each of the components of the bid-ask spread analyzed in Chapter 3, namely: (i) order processing costs or liquidity providers’ rents, (ii) inventory holding costs, and (iii) adverse selection costs.

Order processing costs. Automation of the order submission process should naturally reduce order processing costs for liquidity providers. For instance, replacing human traders with machines reduces the variable labor cost of liquidity provision. Indeed, the development of new trading algorithms involves a fixed cost (required to design and code algorithms), which can be amortized over a large number of trades at negligible variable costs.

Also, recall from Chapter 3 (Box 3.1) that order processing costs may also include liquidity providers’ rents from market power. Algorithmic trading can be expected to intensify competition between liquidity providers and therefore reduce such rents. For instance, consider the arrival of a market order that depletes the limit order book for a stock and thereby increases the bid-ask spread.

By receiving quicker feedback on the state of the limit order book, processing this information faster and automating order submission, liquidity providers can react more quickly to this situation by undercutting posted quotes and thus accelerate the reversal of the bid-ask spread to its competitive level (Cordella and Foucault 1999; Bongaerts and Van Achter 2021). Moreover, as explained in Chapter 6, traders whose limit order has time priority at a given price obtain larger rents. By reacting fast to changes in the state of the limit order book, high-frequency liquidity suppliers can capture these rents.

Inventory holding costs. As explained in Chapter 3, inventory holding costs are smaller for securities in which liquidity providers can unwind their positions more quickly (see, for instance, the discussion of Figure 3.7). High-frequency market makers do so by using very fast access to information on quotes posted in different markets. For instance, consider a high-frequency market maker in Unilever, a stock that trades on various electronic trading platforms in Europe, including Euronext and Cboe. A market maker who fills a sell market order for Unilever on Euronext can immediately place a sell limit order on Cboe and thus turn around his position quickly, possibly faster than if he had traded only on Euronext, especially if Euronext and Cboe attract different types of buy side investors.

Figure 9.1, drawn from Menkveld (2013), presents evidence consistent with this idea. It shows the change in the cumulative positions of a high-frequency market maker in Unilever stock on Euronext and Chi-X (now Cboe) and the sum of these two cumulative positions (i.e., the net change in the market maker's aggregate inventory position), over 200 days. The figure shows that as the market maker buys shares of Unilever on Euronext, he sells shares on Cboe. As a result, his aggregate position fluctuates around zero and his inventory risk exposure never lasts long. More generally, computerized trading enables high-frequency market makers to manage the risk of their positions more efficiently and thus reduce their inventory holding costs (Ait-Sahalia and Sağlam 2017).

Adverse selection costs. Automation of the order submission process also enables traders, in particular high-frequency traders, to react very fast to new information. This possibility has an ambiguous effect on adverse selection costs. To see why, suppose that good news about a stock arrives. It will take some time (from a few milliseconds to minutes) for liquidity providers in this stock to impound the news in their quotes by repricing their limit orders. In the meantime, some traders can submit buy market orders to trade at advantageous prices, that is, against sell limit orders whose prices have not been updated yet. These trades are profitable for the buyers but are costly for sellers. Thus, news arrivals exposes liquidity suppliers to the risk of being adversely selected: they are more likely to sell than buy when news is good (as in our example) and vice versa when news is bad.

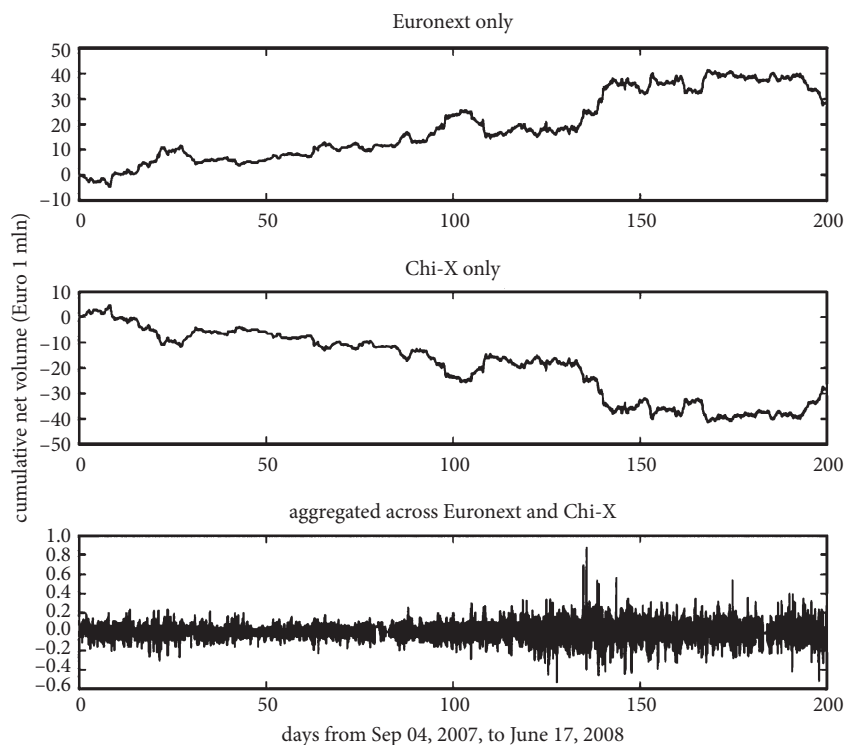


Figure 9.1 Net change in the aggregate inventory (in euros) of a high-frequency dealer in Unilever from September to June 2008.

Source: Menkveld (2014).

This source of adverse selection (referred to as “picking off,” “sniping,” or “free option” risk) is due to differences in traders’ effective reaction time to news (possibly due to differences in technological investments or “jitters,” i.e., randomness in communication times) rather than to differences in their ability to obtain superior information or data processing skills.⁷ This is not specific to modern markets. For instance, around 1835 the reporter Daniel Craig started selling advance access to news from Europe arriving in the U.S. by steamships. Steamships stopped in Halifax, allowing Craig’s agents to read and deliver the news in Boston through carrier pigeons before the steamships arrived there. Craig charged \$500 for each hour of advance access to the news. More recently,

⁷ It is also due to liquidity providers’ obligation to post firm quotes. In some markets, dealers have the possibility to adjust their quotes after receiving an order. This option, called “last look” in currency markets, considerably reduce the risk of trading at stale quotes. It may however create room for opportunistic pricing by dealers, as recent controversies in currency markets show. See, for instance, “The role of last look in foreign exchange markets,” Norges Bank Investment Management, 2015 for further discussion.

day traders in the 1990s took advantage of the automated execution of small market orders on Nasdaq to pick off dealers who were slow to update their quotes after news arrival (see Foucault, Röell, and Sandas 2003). What is novel is the time scale at which this game is played: high-frequency traders react to news in a few milliseconds, rather than in a matter of hours or minutes.

As an illustration, Figure 9.2 (from Brogaard, Hendershott, and Riordan 2014) describes the evolution of stock returns for a sample of Nasdaq stocks over the ten seconds following the arrival of positive macro-economic announcements in the U.S. (left-axis; solid curve) and the net order flow from high-frequency traders, measured by the difference between the dollar volume of buy and sell market orders over the same period (right axis; dashed dotted curve). The figure shows that it takes about 10 seconds on average for stock prices to adjust to their new level after positive announcements. Immediately after the arrival of positive news, HFTs submitting market orders are net buyers of Nasdaq stocks: in about 2 seconds, their net buy is worth on average 2 million dollars. These trades are filled against stale sell limit orders, oftentimes posted by other high-frequency trading firms, who have been slow in cancelling their quotes. This fact is shown by the dotted line, which plots the net dollar difference between buys and sales of high-frequency traders whose limit orders are executed over the 10 seconds following news arrivals. This difference is negative, reflecting the fact that these traders sell more shares than they buy after positive news announcements. As these announcements are followed by an increase in prices on average, these traders are adversely selected.

Interestingly, Figure 9.2 also suggests that the speed game is largely played among high-frequency traders at the expense of each other (the gains of fast traders picking off stale quotes are obtained at the expense of fast liquidity suppliers).⁸ It may however negatively affect all market participants because it leads fast dealers to charge larger bid-ask spreads than they would in the absence of the risk of being picked off, as shown below in Section 9.4.

The exposure to the risk of being picked off decreases if liquidity suppliers become faster in updating their quotes in case of news arrivals *relative* to traders submitting market orders when news arrives (and increases in the opposite case). The effect of high-frequency trading on this risk and its effect of adverse selection costs is therefore ambiguous. If high-frequency trading enables liquidity suppliers to be relatively faster, it should reduce their exposure to the risk of being picked off and through this channel reduces adverse selection. If instead automation makes high-frequency traders faster in picking off stale limit

⁸ In Figure 9.2, the aggregate losses of fast liquidity suppliers exceed the aggregate gains of fast liquidity demanders. This suggests that fast liquidity suppliers' quotes are also picked off by traders who are not classified as high-frequency traders.

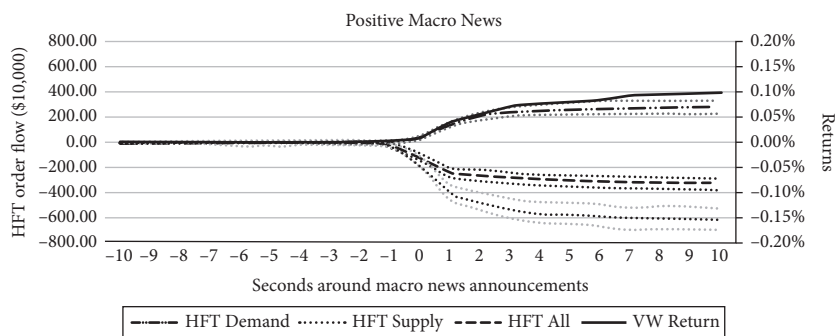


Figure 9.2 Stock prices and trades in Nasdaq stocks following macro-economic announcements.

Source: Brogaard, Hendershott, and Riordan (2014).

orders, it increases liquidity suppliers' exposure to the risk of being picked off and raises adverse selection costs.

Our previous example considered the case of news arrival. It is worth stressing that high-frequency traders do not even need to observe news to pick off stale quotes. Consider, for instance, the case of the E.mini Nasdaq-100 futures and the QQQ Exchange Traded Fund (ETF), which tracks the same index. These two securities are indexed on the same asset. However, liquidity suppliers in each market are distinct (the Nasdaq-100 futures trade on the Chicago Mercantile Exchange in Chicago and the QQQ ETF trade on the Nasdaq data center in Carteret, NJ). When news arrives regarding Nasdaq-100 stocks, these liquidity suppliers will therefore not update their quotes at exactly the same moment, except by chance. Such asynchronous updates of prices in related markets can create very short-lived arbitrage opportunities.

In exploiting these opportunities, high-frequency arbitrageurs create adverse selection for liquidity suppliers in the market whose quotes lag behind (see Budish, Cramton, and Shim 2015). For instance, if liquidity suppliers in the E.mini futures raise their quotes first after good news, fast arbitrageurs can pick off stale sell limit orders in the ETF market while submitting sell market orders in the E.mini futures to hedge their position. Even though these arbitrage trades do not exploit information about subsequent price changes (arbitrageurs do not even need to anticipate them to lock in a profit), they raise adverse selection costs for liquidity providers. Importantly, not all arbitrage opportunities are due to asynchronous reactions of quotes to news arrival and therefore a source of adverse selection. As explained in Chapter 11, even at high-frequency, some are also due to transient demand shocks arising from noise traders: in exploiting these opportunities, arbitrageurs implicitly act as liquidity suppliers and therefore improve liquidity rather than impair it (see Foucault, Kozhan, and Wah 2017).

In sum, the net effect of algorithmic and high-frequency trading on adverse selection costs is theoretically ambiguous. Hence, unsurprisingly, empirical

findings regarding this effect are equally ambiguous. For instance, Chakrabarty et al. (2020) consider the SEC ban on unfiltered market access (so-called “naked access”), requiring brokers to verify that their clients’ orders are compliant with credit and capital thresholds. This verification takes time and therefore reduces the speed with which traders can react to market events. As this ban did not affect traders registered as broker-dealers, it most likely affected high-frequency traders that do not specialize in market making. Consistent with the view that fast trading by HFT can raise adverse selection costs, Chakrabarty et al. (2020) find that the unfiltered access ban is associated with a significant reduction in adverse selection costs and accordingly a drop in trading costs. Similarly, Shkilko and Sokolov (2020) show that rainfalls, which slow down the microwaves transmission of information between Chicago and New York, reduces adverse selection costs in ETFs traded in New York (see Box 9.1 for a description of their methodology). Consistently with these results, Foucault, Kozhan, and Wah (2017) find that a technological innovation that raises arbitrageurs’ relative speed in the foreign exchange market results in an increase in adverse selection costs. All these findings suggest that slowing down arbitrageurs or directional traders exploiting lags in price adjustment across markets reduces adverse selection.

In contrast, Hendershott et al. (2011) find that the introduction of Autoquote on the NYSE (which allows liquidity suppliers to become more quickly informed about changes in the limit order book) is associated with an improvement in liquidity. Similarly, Brogaard et al. (2015) show that market makers become significantly better at avoiding adverse selection after upgrading their co-location with Nasdaq OMX.

In reality, algorithmic and high-frequency trading affect all costs of liquidity provision. Thus, one might find empirically that high-frequency trading improves liquidity because its benefits more than offsets its costs. Thus to better understand how algorithmic and high-frequency trading affects liquidity, it is important to separately analyze its effects on the various components of illiquidity for instance using techniques for decomposing bid-ask spreads described in Chapter 5.

Box 9.1 Rainfall and Liquidity

Various methods are used to transmit information between the Chicago Mercantile Exchange (CME, based in the Chicago area), where investors can trade futures on equity indexes and the NYSE and Nasdaq trading platforms, in New Jersey, where investors can trade Exchange Traded Funds (ETFs). The fastest consists in using microwave networks, which transmit information 30% faster than fiber cable networks (cutting transmission times

from 6.5 milliseconds to 4.5 milliseconds). Users of these networks have therefore a speed advantage in accessing information on quote updates and trades relative to non-users. However, this advantage is reduced when it rains or snows on the transmission path between Chicago and New-York because bad weather conditions slow down information transmission via microwave networks.

Shkilko and Sokolov (2020) shows that during precipitation episodes, effective bid-ask spreads on ETFs drop by about 2.5% and price impacts by about 3.8%. In other words, events that reduce the speed differentials between fast and slow traders mitigate adverse selection costs for liquidity suppliers and thereby improve liquidity.

Last, it is worth stressing that algorithmic trading reduces search costs. As explained in Chapters 2 and 7, equity markets in North America and in Europe are very fragmented. FX and bond markets are even more so given that OTC markets are inherently fragmented (e.g., there are more than 75 trading venues in FX markets; see BIS (2019)). In such an environment, investors or their brokers must decide how to split their orders across trading venues in a way that minimizes their total trading costs. In this sense, they must “search” for liquidity and using algorithms (e.g., smart order routing systems) minimize the time and cognitive costs of this search.

9.3.2 Price efficiency and Informativeness

As explained in previous chapters, another dimension of market quality is the speed at which asset prices impound information about fundamentals. The activity of high-frequency traders can be expected to raise this speed, as these traders are very quick in (i) taking advantage of arbitrage opportunities (thus eliminating them) and (ii) reacting to the arrival of new public information, either by updating quotes to reflect it or by picking off stale quotes (thereby pushing prices in the direction of new information). Intuitively, this behavior should contribute to make prices more efficient, namely, closer to a random walk (see Chapter 3). Empirical findings support this conjecture. Chaboud et al. (2014) find that the introduction of algorithmic trading on foreign exchange trading platforms significantly reduced the duration of arbitrage opportunities in foreign exchange markets. Moreover, Brogaard et al. (2014) find that high-frequency traders’ orders reduce the noise in prices, making them closer to

a random walk. Similarly, Conrad et al. (2015) find a positive cross-sectional relationship between the number of quote updates and price efficiency for U.S. stocks.

In addition, algorithmic and high-frequency traders can exploit information about future price movements or future trades contained in market data. For instance, as explained in Chapter 3, market orders can be seen as noisy signals about investors' private information. Combining computing power with access to vast amounts of market data (e.g., trades in different trading venues involving assets with correlated fundamentals), algorithmic and high-frequency traders can filter out the noise in such data more efficiently than other market participants. In this way, they can "piggyback" on private signals about fundamentals produced by other traders (e.g., hedge funds). This is especially the case when the latter split their orders over time. Doing so, they leave footprints in transactions data, which act as signals for other algorithmic traders.

Korajczyk and Murphy (2018) and Menkveld and van Kervel (2019) provide empirical findings supporting this scenario. Their data allow them to identify episodes of different durations (ranging from a few hours to multiple days) during which institutional investors executed large orders. They observe that high-frequency traders in their samples initially provide liquidity to institutional investors (e.g., place sell orders when institutional investors buy) but eventually trade in the same direction as institutional investors. This behavior is consistent with the existence of a learning phase during which high-frequency traders detect the presence of an institutional investor executing a large order, followed by an exploitation phase in which they trade in the same direction. Interestingly, this behavior is detectable only for trades that appear to be information-based, in line with the idea that high-frequency traders extract information about institutional investors' private signals from their orders.

This behavior has two effects. First, it raises trading costs for institutional investors. Second, it accelerates the speed at which institutional investors' informational advantage decays. Both effects reduce institutional investors' profits from trading on private information and therefore their incentive to produce fundamental information (e.g., through careful analysis of firms' financial reports and prospects). Thus, even though algorithmic or high-frequency trading make prices more efficient, it might reduce the overall production of information and thereby asset price informativeness (see Dugast and Foucault 2018). Consistent with this possibility, Weller (2017) finds that earnings announcements have a stronger impact on the price of stocks with more algorithmic trading, suggesting that the information in these announcements is less frequently anticipated by the market, that is, there is less information production about future earnings in these stocks.

9.3.3 Market Manipulation

Market manipulation is defined by regulators as an activity that artificially changes the supply or demand for a security to affect its price. Actions of this type are not new. For instance, “pump and dump” schemes (spreading false positive rumors about a stock), “wash trades” (the simultaneous entry of buy and sell orders offsetting each other by the same trader or his/her affiliates) or “banging the close” (moving the price of a derivative asset to make profit on a position in the underlying asset) are manipulative practices that existed before the advent of algorithmic trading. However, the use of machines to place orders has raised concerns about new forms of market manipulation, based on quick entry and cancellations of orders. We discuss two examples below.

Spoofing and layering. Spoofing and layering consists in placing and quickly cancelling limit orders on one side of the limit order book for an asset to convey the false impression that the demand or supply for a stock is stronger than it really is. Some algorithmic and high-frequency traders have been accused of market manipulation via such schemes and the number of market manipulation cases brought to courts by regulators (e.g., the SEC and the CFTC in the U.S.) has steadily increased since 2010. For instance, JPMorgan agreed to pay a \$920 million fine in 2020 for spoofing by its traders in metal and Treasury markets from 2008 to 2016 (see “US regulators step up battle with spoofing,” *Financial Times*, October 10, 2020).

As an example, consider an algorithm (“the manipulator”) quickly submitting a series of sell limit orders in a futures market (say, the E.mini futures on the S&P 500) for 1,000 contracts at prices ranging from the best ask price upward, without real intent to sell the futures. Other algorithms may almost instantaneously interpret this change in the state of the book as reflecting weak demand and therefore offer to sell at lower prices. The manipulator can then quickly cancel his own sell limit orders and buy the stock at a better price than he could have otherwise obtained.

By giving the impression of an artificially weak demand, the manipulator influenced the price at which he can buy the futures. Clearly, if successful, this practice defrauds the manipulator’s counterparties and temporarily pushes prices away from fundamental values, thereby impairing price discovery.⁹

Over time, other algorithms may adapt to the risk of spoofing or may find a way to trade at the manipulator’s disadvantage. For instance, in the previous example, algorithms seeking liquidity (e.g., to execute a large buy order) might

⁹ Using Canadian data, Brogaard, Li, and Yang (2022) provide evidence that spoofing harms market liquidity and price discovery. Williams and Skryzpack (2020) presents a model in which spoofing arises in equilibrium, using an extension of Glosten and Milgrom (1985).

take advantage of the attractive sell limit orders submitted by the manipulator by quickly submitting buy market orders before the manipulator cancels them.

Box 9.2 “Make Sure to Always do Your Gravy in Large Size”

In October 2014, in the first market manipulation case involving a high-frequency trading firm in the U.S., the SEC sanctioned a New York City-based high-frequency trading firm for quickly placing large orders, using an algorithm called “Gravy,” with the intent of affecting the closing prices of stocks listed on Nasdaq.¹⁰

On Nasdaq, traders can submit limit and market “on-close orders” that execute at the close of the market. The closing auction price is set at 4:00 p.m. to maximize the number of shares traded at the close under the constraint of not deviating too far from the last price in the continuous session, to reduce volatility.

A few minutes before the closing auction, Nasdaq disseminates information (the “Net Order Imbalance indicator”) about the imbalance between the volume of buy and sell on-close orders at the current market price. Based on this information, market participants can react by entering “Imbalance-Only-Orders” on the opposite side of the imbalance. For instance, a positive imbalance for a stock signals that the demand exceeds the supply for that stock at the close. Thus, market participants anticipate an increase in the price, which encourages them to simultaneously place buy orders in the continuous auction and offsetting sell orders in the close auction. Such a strategy enables traders to profit from the difference between the price at which they accumulate their position in the continuous auctions and the price at which they offset it in the call auction (in a way similar to the optimal behavior of dealers in the presence of anticipated shocks, considered in Chapter 3). In principle, this behavior works to make the price in the continuous auction closer to the closing price.

This strategy is risky, however, unless traders can guarantee the execution of their Imbalance-Only-Order with a high degree of confidence. To do so, one can very quickly place an Imbalance-Only-Order as soon as the Imbalance message is released to gain time priority in the closing auction while progressively accumulating an offsetting position in the continuous auction, particularly in the few milliseconds before the close. The “gravy algorithm” was designed to do so, while making sure that the high-frequency

¹⁰ See <https://www.sec.gov/news/press-release/2014-229> for more information on this case.

trader using it would account for a very large fraction of the trading volume in the milliseconds preceding the auction, as revealed by internal e-mails exchanges between traders such as “make sure we always do our gravy in large size.”

This latter feature (a large size trade) was intended to push the price of the auction in a favorable direction for the trader. It is the one that triggered a regulatory sanction because of the high-frequency trading firm’s explicit intent to affect closing prices. In addition, the SEC argued that by accumulating very large positions, this firm had become a dominant player with significant market power.

Quote Stuffing. This is a strategy that entails submitting a large number of buy and sell orders and cancelling them almost instantaneously. This practice is viewed as a form of market manipulation because it can create a false impression of trading activity in a stock. Moreover, a sudden burst of order submissions delays order processing and the reporting of trade information to all market participants, which creates arbitrage opportunities for investors aware of these delays.¹¹ Hence, some market participants may deliberately swamp platforms with messages (quotes and cancellations) solely to manipulate the tape (the quote and trade information reported to other participants). Egginton et al. (2016) find that quote stuffing events (defined as extreme bursts in the number of quotes submitted per minute) are frequent in U.S. equities markets (on average they count 125 such events per day in their sample), short (more than 70% last less than one minute) and pervasive (74% of all U.S. equities are affected). Moreover, they find that these events are associated with increases in measures of illiquidity (e.g., effective bid-ask spreads) and volatility.

9.4 Investment in Speed as an Arms Race

Huge investments in infrastructure have been made to respond to high-frequency traders’ demand for trading speed (see Section 9.2). These systems are not built by high-frequency trading firms themselves but by third parties (trading platforms or others) who sell access to their infrastructure to

¹¹ See “SEC probes canceled trades,” *Wall Street Journal*, September 1, 2010, regarding the case of Procter & Gamble stock on April 28, 2010. On that day, a burst of limit orders in this stock was transmitted to the NYSE shortly before 11:48 a.m. and cancelled almost immediately, triggering a several-second outage of the NYSE reporting system, sufficient to create an apparent arbitrage opportunity between the prices of Procter & Gamble on the NYSE and the competing BATS platform.

high-frequency traders. However, the cost of these investments is recouped by charging fees to HFTs, which are paid out of their trading revenue.

The demand for trading speed has often been compared to an arms race. In this section, we analyze traders' incentives to become fast (invest in speed) and explain why the race to be fast is indeed, in some respects, comparable to an arms race. To this end, we present a stylized four-period model. This model is similar in many ways to the model of trading with asymmetric information presented in Chapter 3, with two key differences: (i) the asymmetric information stems from differences in the moment at which traders observe public information rather than from private information and (ii) traders can choose the speed at which they react to public information.

Figure 9.3 presents the timeline of the model, which considers the market for a risky asset whose value v can be $v^H = \mu + \sigma$ or $v^L = \mu - \sigma$ with equal probabilities. At the initial date ($t = 0$), traders decide whether to invest in a technology that enables them to react quickly to public information. Then at $t = 1$ a risk neutral dealer posts bid (b) and ask (a) quotes, and at $t = 2$ public information may become available and trade occurs. At the final date ($t = 3$) the asset's value v becomes known, irrespective of whether public information has been revealed at the trading stage.

Denoting the dealer's bid-ask spread by S , the ask and bid quotes posted at $t = 1$ are:

$$a = \mu + \frac{S}{2} \quad \text{and} \quad b = \mu - \frac{S}{2}. \quad (9.1)$$

In choosing these quotes, the dealer anticipates that one of two possible events can occur at the trading date $t = 2$: with probability $1 - \pi$, a liquidity trader submits, with equal probabilities, a buy or a sell order for one share; alternatively, or with probability π , public information (news) is released about the value v of the asset. Such public information might be revealed via announcements by firms, news reports in financial media, discussions in social media, etc.

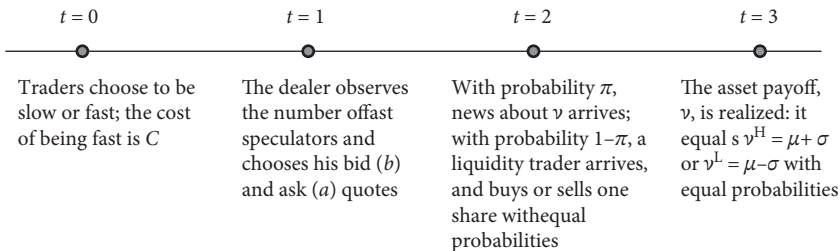


Figure 9.3 Investment in speed as an arms race: time line.

Indeed, some firms (like Ravenpack and Thomson Reuters) specialize in selling trading signals (“buy” or “sell”) to investors based on such announcements. They typically release these signals a few milliseconds after new information arrives (see Keim, Massa, and von Beschwitz 2019), enabling traders subscribing to their services to quickly submit orders based on these signals. For simplicity, we assume that the public signal released at $t = 2$ about the firm is perfect (it reveals v). We first show in Section 9.4.1 how an increase in speculators’ speed of reaction to news raises adverse selection costs for the dealer and therefore the equilibrium bid-ask spread. Then, in Section 9.4.2, we study how traders make investments to be fast.

9.4.1 Trading Speed and Adverse Selection

Now suppose that the dealer’s bid and ask quotes at date $t = 1$ are such that $v^L < b$ and $a < v^H$ (i.e., $S < 2\sigma$). In this case, if public information is released and the dealer does not cancel his quotes, some speculators can trade profitably at the expense of the dealer. For instance, if public information indicates that the value of the asset is high, a speculator can swiftly place a buy market order. His order executes at price a and the speculator makes a profit equal to $v^H - a = \sigma - S/2$, at the expense of the dealer.

This loss is similar to an adverse selection cost for the dealer (see Chapter 3). When he posts quotes at $t = 1$, the dealer must account for the possibility that he might trade, at a loss, with investors who observe public information about the asset value, v , at $t = 2$ before he can cancel (or refresh) his quotes.

However, there is an important difference with the situation analyzed in Chapter 3, where informed traders were assumed to possess private information unavailable to other participants, including dealers. In this case, instead, speculators exploit public information, that is, information in principle available to all market participants. Yet, in a world where traders can react very fast to public information, the frontier between public and private information becomes blurred. In reality, traders do not get the chance to react to public information simultaneously and therefore those who can do so before others have an information advantage, as if they were privately informed.

This asynchronicity in the reaction of traders to news arrival creates a particular form of adverse selection for liquidity providers, which effectively is the “picking off risk” or “sniping risk” described in Section 9.3.1. The exposure of dealers to this risk depends on how fast they react to public information *relative* to other traders. We denote by l_i the probability that trader i “wins the race,” i.e., is the first to react when public information arrives at $t = 2$, where $i = 0$ denotes the dealer and $i = 1, \dots, N$ denotes a speculator.

The expected profit of the dealer (i.e., trader 0) is:

$$\Pi_0(S) = (1 - \pi) \frac{S}{2} - \pi(1 - l_0) \left(\sigma - \frac{S}{2} \right). \quad (9.2)$$

The first term in expression (9.2) is the dealer's realized profit ($S/2$) from trading with a liquidity trader times the likelihood of this event. In this trade, the dealer earns half the bid-ask spread (because he either buys or sells one share with equal probability). The second term is the expected dealer's loss if (i) public information arrives at $t = 2$ and (ii) the dealer is not fast enough to cancel his quotes so as to avoid being picked off. The likelihood of this event is $\pi \times (1 - l_0)$. The bid-ask spread, S^* , such that the dealer breaks even ($\Pi_0(S^*) = 0$) is:

$$S^* = \frac{2\pi(1 - l_0)}{1 - \pi l_0} \sigma. \quad (9.3)$$

We refer to S^* as the “competitive bid-ask spread,” in the sense that it cannot be profitably undercut by another dealer operating at the same speed. It increases with the likelihood of the dealer being picked off, $1 - l_0$, as such an increase raises the adverse selection for the market maker. In the limiting case where the dealer is picked off with certainty ($l_0 = 0$), we obtain the same expression for the bid-ask spread as in the basic adverse selection model presented in Chapter 3: see equation (3.12).

Given the dealers' bid-ask spread S , each speculator's expected profit is:

$$\Pi_i(S) = \pi l_i \left(\sigma - \frac{S}{2} \right), \quad \text{for } i \geq 1, \quad (9.4)$$

that is, the profit earned by the speculator, $(\sigma - S)/2$, times the probability that (i) public information arrives at $t = 2$ and (ii) speculator i is the first to observe it and to place a market order to pick off the dealer, before the latter cancels his quotes: the joint probability of (i) and (ii) is $\pi \cdot l_i$. Hence, when the dealer quotes the competitive bid-ask spread, we obtain the expected profit of speculator i by substituting S^* for S in equation (9.4):

$$\Pi_i(S^*) = \pi l_i \left(\sigma - \frac{S^*}{2} \right) = \frac{\pi(1 - \pi)}{1 - \pi l_0} l_i \sigma, \quad \text{for } i \geq 1. \quad (9.5)$$

Thus, a speculator's expected profit increases with his likelihood of being first to react to the news l_i , because this makes him more likely to trade. It also increases with the volatility of the asset payoff, σ , as this increases the value of the information that the speculator can exploit by trading. Instead, the likelihood of news (π) has an ambiguous effect on the speculators' expected profit. On

the one hand, more frequent news arrival increases it by raising the frequency of his trading opportunities; on the other hand, it lowers it by increasing the competitive bid-ask spread.

9.4.2 Is Investment in Trading Speed Excessive?

So far, trader i 's probability of winning the race, l_i , has been treated as a parameter. But in fact traders may invest in technology to raise their probability of winning the race. To study this investment, consider a simple specification where trader i chooses his "speed" Y_i , and faster traders have a comparative advantage in winning the race, so that the likelihood l_i that trader i is the first to react to the arrival of public information is:

$$l_i(Y_i; Y_{-i}) = \frac{Y_i}{Y_0 + Y_1 + \dots + Y_N} \quad \forall i, \quad (9.6)$$

where Y_{-i} is the vector of all traders' speeds, except trader i . Thus, trader i can increase his likelihood of being the first to react to public information by becoming faster, i.e., by increasing Y_i , but even so he cannot be sure to win the race.¹² This makes sense because in reality a myriad of events outside the traders' control can affect the transmission of their orders to an exchange, so that by chance orders submitted by a slow trader might be executed before faster traders' orders.

For simplicity, we assume that traders can choose between only two possible speed levels, slow or fast. We denote these levels as Y (slow) or $\delta \cdot Y$ (fast), where $\delta > 1$. That is, a fast trader reacts " δ times faster" to events than a slow trader. We initially assume that the dealer is slow (case 1), and then turn to the case in which the dealer can choose to be fast (case 2).

Case 1. Equilibrium with slow market makers

Suppose that there are $F \leq N$ fast speculators and thus $N - F \geq 0$ slow ones. In this case, the dealer's likelihood of being first to update his quotes in response to public information is:

$$l_0 = \frac{1}{1 + N + (\delta - 1)F}. \quad (9.7)$$

¹² This functional form would emerge naturally if the time it takes trader i to react to public information arrival has an exponential distribution with intensity Y_i and reaction times are independent across traders. It is used by Foucault, Röell, and Sandas (2003) and Foucault, Kozhan, and Tham (2017) to analyze a similar problem.

Thus, the chance that dealer manages to move first and avoids being picked off is decreasing in the number of speculators N and, for given N , in the number of fast speculators F active in the market.

Substituting l_0 from equation (9.7) into equation (9.5), we obtain each speculator's expected profit as a function of the number of fast speculators:

$$\Pi_i^j(F) = l_i^j \frac{1 + N + (\delta - 1)F}{1 - \pi + N + (\delta - 1)F} (1 - \pi)\pi\sigma, \text{ for } i \geq 1, \quad (9.8)$$

where l_i^j is the likelihood that speculator i of type $j \in \{slow, fast\}$ is first to react to news. For a slow speculator this probability is, from expression (9.6),

$$l_i^{slow}(F) = \frac{\Upsilon}{(N + 1 - F)\Upsilon + \delta F\Upsilon} = \frac{1}{1 + N + (\delta - 1)F}, \quad (9.9)$$

while for a fast speculator it is:

$$l_i^{fast}(F) = \frac{\delta\Upsilon}{(N + 1 - F)\Upsilon + \delta F\Upsilon} = \frac{\delta}{1 + N + (\delta - 1)F}. \quad (9.10)$$

Thus, a fast speculator is δ faster than a slow one and is therefore δ times more likely to win the race among speculators. For this reason, the expected trading profit of a fast speculator is δ times larger than that of a slow speculator:

$$\Pi_i^{fast}(F) = \delta \Pi_i^{slow}(F). \quad (9.11)$$

The expected profit of all speculators decreases with the total number of speculators, N , and for given N it decreases with the number of fast ones, F , because an increase in the number of fast speculators lowers any trader's chance of winning the race. Indeed, revenues from high-frequency trading have sharply declined over time, from over \$9 billion in 2009 to about \$2 billion in 2016.¹³ This drop may reflect entry of new fast trading firms (higher F), as well as a reduction in volatility (lower σ).

At date 0, speculators simultaneously choose whether to invest or not in a fast trading technology (e.g., more powerful computers or access to market data via colocation or microwave transmission), whose cost is C . The outcome of this choice by speculators is observed by all at date 1: hence, the market maker observes F when he sets his quotes.

¹³ See "How high-frequency trading hit a speed bump," *Financial Times*, January 1, 2018.

The situation in which all speculators are fast (i.e., $F = N$) is a Nash equilibrium if each speculator is better off investing (being fast) than not investing when he expects (i) all $N - 1$ other speculators to be fast and (ii) the bid-ask spread to be equal to that charged by the market maker when all other speculators are fast. A necessary and sufficient condition for this equilibrium to exist is that no speculator can be better off by choosing to be slow unilaterally at date 0, when all others are fast. This requires:

$$\Pi_i^{fast}(N) - C \geq \Pi_i^{slow}(N - 1). \quad (9.12)$$

As $l_i^{slow}(N - 1) = \frac{1}{2 + \delta(N - 1)}$ and $l_i^{fast}(N) = \frac{\delta}{1 + \delta N}$, this condition becomes:

$$(\delta - 1) \frac{1 - \pi + \delta(N - 1)}{(1 - \pi + \delta N)(2 - \pi + \delta(N - 1))} (1 - \pi)\pi\sigma \geq C, \quad \text{for } i \geq 1. \quad (9.13)$$

The left-hand side of this inequality is increasing in the relative speed advantage of being fast, δ , and the volatility of the asset, σ , and decreasing in the number of speculators, N . Thus, an equilibrium in which all speculators are fast is more likely to obtain when the speed advantage of the fast trading technology is large, asset volatility is high and there are not too many speculators trying to pick off the dealer, so that speed confers a non-negligible advantage.

Arms Race between Speculators. The previous argument shows that under certain conditions all speculators find it optimal to invest in high-frequency trading technologies when they expect others to do so. Is this a desirable outcome from the viewpoint of speculators? Not necessarily: the availability of such technology can lead to over-investment, i.e., expenses that speculators would prefer to avoid if they could precommit to do so. This is the case when speculators are better off when they are all slow ($F = 0$) than when they are all fast ($F = N$), that is when:

$$\Pi_i^{fast}(N) - C - \Pi_i^{slow}(0) = \frac{(\delta - 1)}{(1 - \pi + N)(1 - \pi + \delta N)} \pi(1 - \pi)^2 \sigma - C < 0. \quad (9.14)$$

It is easily seen that this condition is always satisfied, even if C is small, when π is sufficiently close to 1. In this case, speculators would be better off if they could commit to being all slow instead of being all fast. The reason is that the fast trading technology has two opposite effects on speculators' expected trading profit. It raises the likelihood that one of them will be able to react to news before the market maker updates his quotes. However, when this happens, a speculator obtains a smaller expected profit because the dealer charges a larger

bid-ask spread to protect himself from faster speculators. When π increases, the second effect strengthens and can eventually dominate the first.

Thus, when conditions (9.13) and (9.14) hold, there is an equilibrium in which all speculators are fast and yet, they would all be better off if they could all commit to be slow.¹⁴ In this case, the equilibrium features overinvestment even from speculators' standpoint. This is one reason why one can compare investment in the high-frequency trading technology to an arms race. Such an investment can ultimately make all speculators worse off after accounting for the cost of investing in speed and the adjustment in bid-ask spreads. Yet, individually each speculator has no incentive to remain slow ("unarmed") if he expects all others to be fast ("build up weaponry"). Figure 9.4 shows that the set of values for δ and C for which conditions (9.13) (the solid curve) and (9.14) (the dashed curve) hold is non-empty (other parameters being fixed at $\pi = 0.2$, $N = 10$, and $\sigma = 10$).

The inefficiency comes on top of the inefficiency of high-frequency trading from the standpoint of liquidity traders, because an increase in the number of high-frequency traders reduces the likelihood, l_0 , that the dealer can cancel his quotes before being picked off, which raises the bid-ask spread by equation (9.3). Thus, the bid-ask spread is always larger when all speculators are fast than when they are all slow.

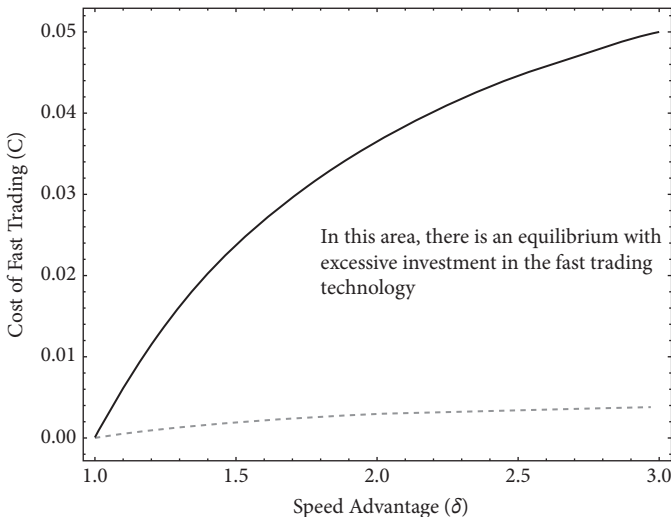


Figure 9.4 Excessive investment in the fast trading technology.

¹⁴ The case in which all speculators are slow is not an equilibrium in this case. Indeed, one can check that if condition (9.13) is satisfied, then a speculator is better off becoming fast if he expects other speculators to be slow, that is $\Pi_i^{fast}(1) - C - \Pi_i^{slow}(0) > 0$.

Not only can there be overinvestment from the viewpoint of speculators and increased trading costs for liquidity traders: there is also always overinvestment from the overall social viewpoint. This is because the “arms race” to pick off stale quotes has no social benefit in the model. The aggregate expected profit of speculators is just equal to the expected trading cost for liquidity traders, because trading is a zero sum game. Thus, for traders as a whole, total expected profit is zero, so that the speculators’ investment in trading speed is just a social loss. Nobel prize winner Paul Krugman voiced precisely this concern about high-frequency trading in a 2009 *New York Times* column:

High-frequency trading probably degrades the stock market’s function, because it’s a kind of tax on investors who lack access to those superfast computers—which means that the money Goldman spends on those computers has a negative effect on national wealth. As the great Stanford economist Kenneth Arrow put it in 1973, speculation based on private information imposes a “double social loss”: it uses up resources and undermines markets.¹⁵

This is exactly what happens in the case in which only speculators invest in fast trading: (i) high-frequency trading uses up resources and (ii) undermines markets (the bid-ask spread increases in equilibrium). It remains however to be checked whether this conclusion still stands when one considers that market makers may also choose to invest in fast trading: this is what we turn to next.

Case 2: Incentives to be fast for market makers.

Differently from the analysis developed so far, we now allow the dealer to invest in the fast trading technology at date 0. To simplify the analysis, we initially suppose that the dealer ($i = 1$) faces competition from a fringe of slow dealers who stand ready to offer liquidity at their competitive bid-ask spread, given by equation (9.3). That is, if the dealer were to charge a larger spread, one of them would step in and post the competitive spread. Otherwise they remain inactive.

Suppose that condition (9.13) is satisfied. Then, if the active dealer does not invest in the fast trading technology, there is an equilibrium in which all speculators choose to invest (see Case 1). In this case, the dealer charges a bid-ask spread equal to:

$$S^* = \frac{2\pi(1 - l_0)}{1 - \pi l_0} \sigma = \frac{2\pi\delta N}{1 - \pi + \delta N} \sigma, \quad (9.15)$$

¹⁵ P. Krugman, “Rewarding bad actors,” *New York Times*, August 2, 2009.

because $l_0 = \frac{1}{1+\delta N}$ when all speculators are fast and the active dealer is slow. So the active dealer just obtains an expected trading profit of zero

If instead the active dealer chooses to be fast, she still charges the same bid-ask spread S^* . Indeed, a smaller bid-ask spread would reduce her expected profit while a larger expected bid-ask spread would be undercut by slow dealers. However, in contrast to case in which she is slow, her likelihood of being able to cancel her quotes before being picked off, is larger than when she is slow: it is $l_0 = \frac{1}{1+N}$ rather than $l_0 = \frac{1}{1+\delta N}$. Thus, the active market maker's expected trading profit is:

$$\begin{aligned}\Pi_0(S^*) &= (1-\pi)\frac{S^*}{2} - \pi(1-l_0)\left(\sigma - \frac{S^*}{2}\right) \\ &= \frac{N(\delta-1)\sigma(1-\pi)\pi}{(N+1)(1-\pi+\delta N)} > 0.\end{aligned}\tag{9.16}$$

It is strictly positive because the dealer's adverse selection cost is smaller when she is fast than when she is slow and, as she is assumed to be the only fast dealer, she can afford to charge a non-competitive bid-ask spread from her own standpoint.¹⁶

However, the active dealer will not want to invest in the fast trading technology if her expected trading profit when she is fast net of the cost of the technology, $\Pi_0(S^*) - C$ is negative, that is, if:

$$\frac{N(\delta-1)\sigma(1-\pi)\pi}{(N+1)(1-\pi+\delta N)} < C,\tag{9.17}$$

Thus, when conditions (9.13) and (9.17) hold (the set of parameters such that this is the case is non empty), there is an equilibrium in which all speculators are fast and the active dealer optimally choose not to be fast, so that the equilibrium is as in case 1.

This analysis highlights an important difference between the decision to be fast for speculators and the decision to be fast for dealers. The former compares the expected profit they can obtain by picking off stale quotes to the cost of being fast. The latter compares the expected profit from liquidity provision (stemming from trades with liquidity traders) with the cost of being fast. This profit is constrained by the extent of competition from other dealers, even the slow ones. This difference between the speculators and the dealer explains why

¹⁶ This situation is akin to a situation in which two firms compete in prices with different unit costs. The firm with the lowest unit cost can charge a price just below the cost of its competitor and earn a profit equal to the difference in cost.

an equilibrium can exist where the former choose to be fast while the latter does not.¹⁷

Competition between fast dealers reduces even more dealers' incentives to be fast. Indeed, when at least two market makers are fast, price competition at date 1 drives their expected trading profit to zero, preventing them to recoup the cost of investing in the fast trading technology. If dealers correctly anticipate this outcome, not investing in the fast trading technology is optimal for the dealers.¹⁸

9.4.3 Selling Information to Fast Traders

The previous model can be interpreted as a model of information acquisition. Paying the cost C at date 0 enables traders to obtain information about the asset payoff at date 1. In fact, trading platforms that provide co-location services, companies that provide quick data feed using high-speed fiber or microwave transmissions or sell high-speed access to newswires are indeed selling information. By charging fees for their services, they capture a fraction of the rents earned by traders subscribing to their services.

The risk of overinvestment in information might not be specific to the type of information obtained by these traders. However, it might be more acute. Compare two cases: an asset manager who hires an analyst to assess a firm's prospect and a trading firm that subscribes to a fast data feed to obtain information on quote updates (or public news) a few milliseconds before other participants.

In the former case, the analyst produces information that would not necessarily be available otherwise. By trading on this information the asset manager generates adverse selection for other market participants but this has the benefit of making the firm's stock price more informative. Hence, the investment made by the manager to obtain private information has social value (see Chapter 12), even though it is a source of adverse selection. In contrast, the information obtained by the trading firm would have been released to market participants in any case and therefore the social return on the cost paid by the trading firm to obtain this information is probably negligible. In the words of Hirshleifer (1971), the trading firm just obtains "foreknowledge" (information about events that in due time will anyway become known to all) while the analyst contributes to information discovery (i.e., produce information that otherwise would never be available).

¹⁷ This is consistent with evidence in Aquilina et al. (2021) who find that the fastest traders in their data tend to submit market orders rather than limit orders.

¹⁸ However, if all dealers are slow, one may find attractive to become fast if condition (9.17) is not satisfied. This reasoning implies that there is no equilibrium in which dealers decide to invest or not invest in the fast technology with certainty (pure strategy equilibrium). One must then consider equilibria in which the investment decision of a dealer is determined using a mixed strategy.

9.5 Algorithmic Trading and Financial Stability

Another concern about algorithmic trading is its impact on the stability of financial markets. This concern stems from the fact that the automation of trading creates new risks or changes existing ones (in addition to increasing the risk of market manipulation, as discussed in Section 9.3.2).

Operational Risks. First, the scale and speed at which algorithms can deploy trading strategies increase operational risks due to malfunctioning of algorithms. The demise of Knight Capital following the deployment of an ill-designed algorithm in 2012 provides a vivid illustration of this type of risk (see Box 10.3).¹⁹ To reduce operational risks, regulators have begun to require that trading algorithms are appropriately checked and tested before going live. For instance, the SEC's regulation Systems Compliance and Integrity Rule (SEC (2013) requires exchanges and traders using computerized trading systems to:

establish written policies and procedures reasonably designed to ensure that their systems have levels of capacity, integrity, resiliency, availability and security adequate to maintain their operational capability and promote the maintenance of fair and orderly markets, and that they operate in the manner intended.

MIFID II (Article 48(1)) in Europe has even more explicit requirements regarding risk controls in place for algorithmic traders. For instance, it requires firms to have a “kill switch” function to completely stop transactions in case an algorithm malfunctions.

Box 9.3 The Demise of Knight Capital

Knight Capital (KC) was one of the largest market making firm in the U.S. until the dramatic events of August 1, 2012. On that day, before the market opening, KC updated its smart order routing algorithms, but by mistake it forgot to disable a particular functionality used to test these algorithms in artificial markets. As a result, its algorithms could not recognize whether

¹⁹ Another example is the failure of the IPO of BATS Global market on March 23, 2012, because of a mishap in the algorithm used by BATS to run the electronic auction for its own IPO. See “IPO software behind BATS failure,” *Financial Times*, March 26, 2012. Ironically, BATS Global Market (now Cboe) is one of the major equity markets, both in the U.S. and in Europe. More generally, exchanges around the world often experience outages due to software malfunctions. Recent examples are discussed in the article “Exchange outages spark demands for action,” *Financial Times*, May 18, 2021.

orders sent for execution were executed or not. Thus, they kept sending million of buy or sell orders in various stocks over the first 30 minutes of the trading day. As a result, KC accumulated large long and short positions in about 150 stocks for a total value of about \$7 billion.²⁰

To avoid bankruptcy due to margin calls on the next day, KC quickly tried to reduce its positions. However, this could be done only at significant discount or markup relative to the purchase price for long or short positions, respectively. Eventually, KC agreed to sell its entire remaining net position to Goldman Sachs at a 5% discount, resulting in a total trading loss of \$460 million. The week after it received a rescue package of \$400 million from six different trading firms and was eventually acquired by GETCO (another high-frequency market maker) in December 2012.

Interestingly, this event had significant effects on the prices and liquidity of stocks bought or sold by KC, as shown by Bogousslavsky et al. (2020). The price of bought stocks first rose (and those sold dropped) very significantly, as predicted by the models considered Chapters 3 and 4. Then, once the glitch became public information and market participants understood that the massive buys and sells were not due to private information, prices started reverting toward the pre-glitch level. But they did so slowly, consistent with models based on inventory risk (see Chapter 3), and it took about one day for the initial price effects to dissipate. Moreover, trading costs for institutional investors who were seeking to trade in the same direction as KC increased significantly for several days.

Market Liquidity Risks. Another potential risk of algorithmic and high-frequency trading is that it increases the likelihood of sudden liquidity dry-ups, that is situations in which liquidity indicators (bid-ask spreads and depth) for a security (or several securities) deteriorate very quickly.

There might be several reasons for this. First, high-frequency market makers control their inventory holding costs by turning around their positions quickly. However, this is possible only if the order flow is not too large and one-sided for too long. If not, high-frequency market makers switch from supplying to demanding liquidity, which results in a sudden evaporation of liquidity and an amplification of price impacts. Another, related, reason is that high-frequency market making firms are more likely to stop their machines in periods of abnormal market conditions (e.g., large unexplained price changes), because their algorithms are trained to operate in normal times or because large price

²⁰ See Bogousslavsky et al. (2021) for a detailed description of this event and its consequences.

movements lead them to reassess the risk of their positions. Last, algorithms might be more prone to “herding” because they are trained to react in the same way to the same signals or are so good at forming predictions that this leaves little disagreement among them in the interpretation of the same data. In this case, high-frequency market makers might decide to curtail or expand liquidity provision in the same situations, triggering extreme ups and downs in liquidity supply.

The U.S. Treasury market crash on October 15, 2014, offers an interesting example of such sudden liquidity dry ups: the yield on the recently issued (“on-the-run”) 10-year Treasury security (an important benchmark for other interest rates) experienced a “Flash rally” (a sudden sharp increase in prices followed by a reversal) between 9:33 and 9:45 a.m., without any apparent cause. Over this period, the 10-year Treasury yield decreased and then increased by about 16 bps. A movement of this size over such a period of time is highly unusual in U.S. Treasury markets, which are among the most liquid in the world.

For this reason, and given the importance of U.S. Treasury markets for the functioning of the world economy, this event was followed by an in-depth investigation by various agencies (including the FED, the SEC, and the CFTC).²¹ The inquiry established that the sharp decrease in yields (increase in Treasury prices) coincided with a sudden decrease in the depth of the limit order book for the 10-year Treasury Bill. Interestingly, at the beginning of the event window (9:33–9:45), the bulk of the liquidity supply came from Principal Trading Firms (PTFs) using quantitative trading strategies rather than market making trading desks. However, as the event unfolded, they reacted by reducing the size of their limit orders (depth at the best quotes) while bank dealers reacted by raising their bid-ask spreads. The inquiry concluded that: “Both actions served as risk management strategies by reducing the number and size of orders that could be executed, and also caused a sharp drop in the supply of liquidity to the market.”

The inquiry could not find a clear catalyst for the Treasury crash. In particular, in contrast to the 2010 Flash crash (discussed below), it could not attribute the initial trigger of the price movement to the arrival of large buy orders. One possibility is that this type of event reflects technological changes that happened over the last twenty years in Treasury markets. In particular, trading in “on-the-run” Treasuries is now done on electronic trading platforms. This automation has triggered entry of PTFs, which now dominate liquidity provision in on the run Treasury markets.

²¹ See “Joint Staff Report: The U.S. Treasury Market on October 15, 2014,” 2015, available at: <https://home.treasury.gov/system/files/276/joint-staff-report-the-us-treasury-market-on-10-15-2014.pdf>.

Another contributing factor are new regulations in the wake of the 2007–8 financial crisis that increased inventory holding costs for traditional liquidity providers (market-making trading desks of large banks). It is possible that PTFs compensated the lower ability of traditional liquidity providers to post tight spreads by using trading strategies that are less exposed to inventory risk in normal times. However, as PTFs are more lightly capitalized, they cannot keep positions for long.

Propagation Risks. Many securities are closely related, for instance stocks that are part of a well-diversified index (say, the EuroStoxx 50), futures on this index and Exchange Index Funds tracking this index. As explained in Section 9.2, algorithmic traders will often seek to quickly take advantage of temporary price discrepancies between these assets. One concern is that in doing so they might transmit non-fundamental shocks in one asset to other assets, possibly triggering large non-fundamental price movements that would not have occurred in the absence of fast arbitrageurs.

The Flash Crash of May 6, 2010, is a good example of this propagation risk. On that day, the Dow Jones Industrial Average experienced its second-largest intra-day point swing ever recorded, falling by 9% in a matter of minutes before rebounding by 5% by the end of the day. The stocks making up the index lost about \$1 trillion in market value in the half hour between 2:30 p.m. and 3:00 p.m., before bouncing back to a level not too far from the initial one. Some individual stocks underwent even more dramatic swings. For instance, Accenture shares fell by over 99%, from \$40 to \$0.01, while Sotheby's shares rose three-thousand-fold, from \$34 to \$99,999.99.

A joint report by the Commodity Futures Trading Commission (CFTC) and the SEC (CFTC-SEC 2010) found that the crash may have been triggered by a single very large sell order in the E-mini S&P 500 index: a large mutual fund sold an unusually large number of futures; first the order exhausted available buy limit orders, and then it precipitated additional sales by high-frequency traders, which spread the crash quickly from the futures market to the stock market.

Figure 9.5 (drawn from the CFTC-SEC report on the flash crash) shows the evolution of the depth of the LOBs of the E-mini futures contract on the S&P 500 index, the SPY Exchange-traded fund and the stocks in the S&P 500 index (averaged across all its individual constituents). Depth is measured as a fraction of the depth available at 1:30 p.m. (the start of the crash). The evaporation of liquidity in all these securities is evident: by 2:45 p.m., the liquidity available in each market was down to just 20% of its level at 1:30. The LOBs for these securities then started replenishing as traders submitted new limit orders. This injection of new limit orders was a factor in the upturn of prices and liquidity observed towards the end of the crash. These dynamics after a large sell order

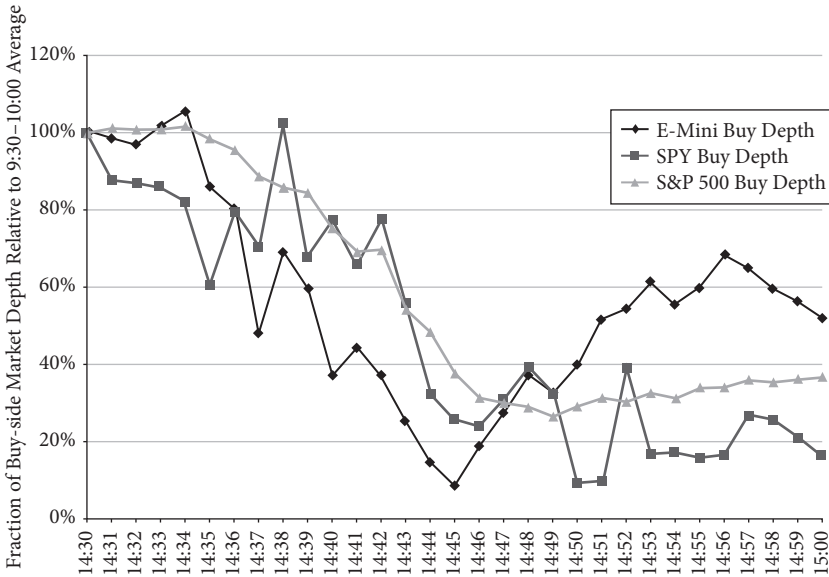


Figure 9.5 Market liquidity of the E-mini futures on the S&P 500, the SPY ETE, and the constituent stocks of the S&P 500 from 2:30 p.m. to 3 p.m. on May 6, 2010.

Source: CFTC-SEC 2010.

(a drop in price and liquidity followed by an upturn) is an extreme manifestation of a more general phenomenon in illiquid markets: buy and sell market orders trigger transient price movements followed by reversals (see Figure 0.2 in the Introduction). Chapter 3 explores the reason for such reversals.

The pathological nature of the flash crash lies not only in the extraordinary magnitude of the collapse in price and liquidity and the subsequent recovery, but also in the fact that it propagated so widely and so quickly across different markets. The reason for this is not completely clear, but algorithmic trading is likely to have been a factor. Indeed, in the face of a drop in the price of E.mini futures relative to the underlying basket of stocks, high-frequency arbitrageurs probably purchased the E.mini futures (and in this way they dampened the price pressure exerted by the large seller) and simultaneously sold stocks in the underlying basket (and this transmitted the initial price shocks to underlying securities). Moreover, directional algorithms may have mistakenly interpreted the drop in the E.mini futures price as a negative signal about the value of underlying stocks, selling them. These actions could have contributed to propagating the E.mini futures crash to underlying stocks, even those that were not constituents of the S&P 500 index. In this way, even though they were not at the origin of the crash, high-frequency traders are likely to have exacerbated the resulting price decline and spread it across markets.

In sum, the automation of trading makes markets more interconnected. While in principle this should contribute to better and faster price discovery, insofar as prices are noisy signals, it also increases the likelihood that traders in one security may react to price movements in another security that are actually triggered by uninformative trades.

Some market participants also claim that “mini-flash crashes,” large and short-lived price movements affecting one or just a few stocks simultaneously, have become more frequent with the advent of algorithmic and high-frequency trading.²² The academic literature on this topic is still scarce. However, existing findings do not support this view.²³ Brogaard et al. (2018) examine high-frequency traders’ behavior for a large sample of Nasdaq stocks during extreme price movements (increases or decreases in returns in the 99.9th percentile of the return distribution in their sample). On average, they find that during these episodes, high-frequency traders’ aggregate net trade is in a direction opposite to the price movement. Thus, rather than amplifying extreme price movements, HFTs seem to dampen them by trading against “the wind.”

9.6 Slowing Down High-Frequency Traders

As explained in previous sections, high-frequency traders can increase the exposure to adverse selection for liquidity providers. Moreover, we have seen in Section 9.4.2 that the technological race by traders to pick off stale limit orders following news arrival or avoid being picked off can lead to overinvestment. Last, the trading strategies of high-frequency and algorithmic traders generate a very high level of cancellations and order resubmissions, which puts a strain on electronic trading platforms and can make financial markets fragile.

Various changes in the design of continuous limit order markets have been proposed to mitigate these problems. These proposals include:

1. **Minimum resting times for limit orders.** According to this proposal, limit orders must remain in limit order books before cancelled for a minimum amount of time. For instance, in June 2009, ICAP introduced a minimum resting time (of 250 milliseconds) for quotes posted on EBS

²² See “Mini Flash Crashes Worry Traders,” *USA today*, May 17, 2011, and “German Bundesbank: High-Frequency Trading Can Worsen ‘Flash Crashes,’” *Wall Street Journal*, October 24, 2016.

²³ Some mini-flash crashes seem to be due to erroneous orders. In 2016, Merrill Lynch agreed to pay a \$12.6 million fine for submitting erroneous large orders (due to bad internal control procedures) and causing mini-flash crashes. See “Merrill Lynch charged with trading controls failures that led to mini-flash crashes,” available at <https://www.sec.gov/news/press-release/2016-192>.

(Electronic Broking Service), an electronic limit order book used in currency markets.

2. **Caps (or fees) on Messages-to-Trade Ratios.** This intervention sets a cap the number of orders (cancellations and resubmissions) relative to the number of trades for a given trader (or charge a fee for exceeding this cap).
3. **Speed bumps.** They consist in delaying the time at which new orders submitted to a trading platform are processed by its matching engine. IEX (“The Investors Exchange”) was the first trading venue to develop this feature using 38 miles of fiber cable coiled in a small box (called the “magic shoe box” by IEX) to delay the arrival of all orders at its matching engine by 350 millionths of a second. Several futures and stock exchanges are now considering to use speed bumps.²⁴ Many (like for instance the Cboe) have proposed asymmetric speed bumps such that incoming market orders face longer delays than cancellations.
4. **Randomized execution times for market orders.** This is a particular form of speed bump in which the time between the moment a market order arrives at an exchange and its entry in the exchange’s matching engine is chosen randomly. For instance, the Toronto Stock Exchange (TSX) introduced such a randomized speed bump on “Alpha” (one of its trading platforms). Alpha delays all incoming market and limit orders by a random time ranging from 1 to 3 milliseconds. However, traders can bypass the delay by paying a fee to the TSX.
5. **Frequent batch auctions.** Budish, Cramton, and Shim (2015) argue that, by design, continuous limit order markets encourage excessive investment in speed because the continuous-time serial processing (i.e., in their sequence of arrivals) of orders creates very frequent arbitrage opportunities (i.e., opportunities to pick off stale limit orders). Therefore, they propose to replace continuous limit order markets with batch auctions taking place at discrete points in time (such auctions would be run at very high-frequency but not continuously). Indriawan et al. (2021) consider a switch from batch auctions to continuous trading by the Taiwan Stock Exchange and find, as proposed by Budish et al. (2015), that this switch triggers an increase in adverse selection for liquidity providers.

Speed bumps, randomized execution times for market orders and batch auctions aim at reducing (in different ways) the speed advantage of fast traders (δ in the model of Section 9.4) to make investment in fast trading less attractive.

²⁴ See “More platforms add speed bumps. Defying high frequency traders,” *Wall Street Journal*, July 29, 2019 and “Futures exchanges eye shift to ‘flash boys’ speed bumps,” *Financial Times*, May 30, 2019.

From equation (9.11), one obtains that the difference between the expected profit of fast and slow speculators (when all dealers are slow) is:

$$\begin{aligned}\Pi_i^{fast}(F) - C - \Pi_i^{slow}(F) &= \\ (\delta - 1)\Pi_i^{slow}(F) - C &= \\ (\delta - 1)\frac{(1 - \pi)\pi\sigma}{(1 - \pi + N + (\delta - 1)F)} - C.\end{aligned}\tag{9.18}$$

Thus, when δ decreases, the difference in expected profit between fast and slow speculators declines and the largest possible number of speculators who choose to be fast in equilibrium (the largest value of F such that the previous expression is positive) must decline as well. In this sense, “slowing down” high-frequency traders is a way to mitigate investment in speed.

Others have proposed that a financial transaction tax (FTT), the so-called “Tobin Tax,” could be a way to curb high-frequency trading and other forms of speculative activities. For instance, the Inclusive Prosperity Act proposed by the 2020 U.S. presidential candidate Bernie Sanders stated that an FTT would “reduce speculation and high-frequency trading that is destabilizing financial markets.” Colliard and Hoffman (2017) study the effects of the introduction of a Financial Transaction Tax on August 1, 2012, on the daily change in net positions (purchases) for stocks traded on Euronext (the French Stock Exchange).²⁵ They find that stocks affected by the French FTT (stocks with a market capitalization exceed \$1 billion) experience a significant drop in trading volume, no change in volatility and a moderate increase in measures of illiquidity. As intraday transactions were exempt from the tax, the FTT did not directly affect high-frequency traders operating on Euronext.²⁶ However, the trading activity of HFTs declined by 35% due to (i) the increase in bid-ask spreads (so that submitting market orders is less profitable for HFTs) and (ii) a decrease in intermediation needs, as trading volume drops. Interestingly, Colliard and Hoffman (2017) also show that market orders submitted by high-frequency traders have the strongest price impact among orders submitted by all market participants (in line with the idea that these traders are well informed). Thus, the drop in their activity reduces adverse selection costs but this is insufficient to offset the increase in bid-ask spreads due to other effects of the tax.

²⁵ In October 2012, 11 E.U. countries committed to the introduction of an harmonized tax on financial transactions.

²⁶ In addition to the Financial Transaction Tax, France imposed a tax of 1 bps on the notional amount of modified and cancelled orders for HFTs with an order-to-trade ratio larger than 5. However, this tax only applied to HFTs based in France and market making activities were exempt. De facto, the most active high-frequency trading firms were therefore exempt from the tax.

9.7 Further Reading

The literature on algorithmic and high-frequency trading is growing fast. Surveys of this literature are provided by Biais and Foucault (2014), Goldstein et al. (2014), O'Hara (2015), Menkveld (2016), Foucault and Moinas (2018), SEC (2020), and Duffie et al. (2022) (Chapter 4).

Theoretical studies. Copeland and Galai (1983) is the first paper to note that limit orders are exposed to the risk of being picked off after news arrival. Foucault et al. (2003), Hoffman (2014), Budish et al. (2015), Foucault et al. (2017), Menkveld and Zoican (2017), and Baldauf and Mollner (2020) analyze theoretically races to pick off stale quotes due to news arrivals. Foucault et al. (2003), and Foucault, Kozhan, and Tham (2017) consider models in which investors can choose the speed to which they react to news or arbitrage opportunities as in the model of Section 10.4. Hoffman (2014) considers the possibility for fast traders to post limit orders in a model of limit order trading (similar to that considered in Section 6.4.1). Budish et al. (2015) analyze how batch auctions can reduce liquidity providers' exposure to the risk of being picked off and reduce incentives to overinvest in speed.

Using a model similar to Budish et al. (2015), Menkveld and Zoican (2017) consider the effect of delays in order processing by trading platforms on bid-ask spreads. Baldauf and Mollner (2020) extend Budish et al. (2015)'s model to a multi-market setting and show that high-frequency traders reduce incentives to produce fundamental information. They then analyze the effects of speed bumps and batch auctions on bid-ask spreads and incentives to produce information.

Other papers assume that high-frequency trades have information about short-term price movements. Biais et al. (2015) study a model in which high-frequency trading enables investors to obtain information about short-term price movements and posted prices in a fragmented market. They show that in equilibrium, the fraction of investors choosing to be fast always exceeds the social optimum. Foucault et al. (2016) extends the Kyle (1985) model when the informed trader has both long-lived private information about the long run payoff of the asset and short-lived private information about public news about the asset payoff. They show that the latter possibility significantly affects the informed investor's optimal trading strategy. Yang and Zhu (2019) consider a two trading periods model similar to Kyle (1985). After the first period, some traders ("back runners") receive a signal about informed investors' trades in the first period and trade on this signal in the second period. They show that the presence of back runners induce the informed investors to play a mixed strategy in period 1 (as in Levine et al. (2001)).

In Cespa and Vives (2022) multiple exchanges control the fraction of speculators who can access a fast trading technology (e.g., pay a co-location fee). In their

model, fast speculators have more trading opportunities (can trade at multiple dates) than slow traders. An increase in the number of fast speculators increases market liquidity and welfare (by intensifying competition in the provision of liquidity supply). Competition induces exchanges to reduce the fee charged for accessing the fast technology and improves welfare. However, it leads to excessive entry (market fragmentation) relative to the first best outcome.

In contrast to the previous studies, Aït-Sahalia and Sağlam (2017) derive the optimal pricing behavior of a high-frequency market maker facing inventory holding costs. They show that his bid-ask spread is smaller on average when the dealer is faster because she can then better manage her inventory risk.

Empirical studies. Several authors study the effect of algorithmic and high-frequency trading on various aspects of market quality. Hendershott, Jones, and Menkveld (2011), Hendershott and Riordan (2013), Brogaard, Hendershott, and Riordan (2014), Brogaard, Hagstromer, and Norden (2015), Conrad, Wahal, and Xiang (2015), and Boehmer, Li, and Saar (2018) provide evidence of a positive effect of algorithmic or high-frequency trading on liquidity and price efficiency. Other studies focus more specifically on the effect of trading on adverse selection costs and find a positive effect of fast trading on adverse selection costs (e.g., Foucault, Röell, and Sandas (2003), Foucault, Khozan, and Tham (2017), Chakrabarty et al. (2020), Shkilko and Sokolov (2020), Aquilina, Budish, and O'Brien (2021), Indriawan et al. (2021)).

Baron et al. (2018), Chordia et al. (2018), and Brogaard and Garriott (2018) estimate high-frequency traders' profits in various environments and study the determinants of these profits. Chao, Chen, and Ye (2018) show that high-frequency traders compete for time priority and are therefore more likely to provide liquidity when the tick size is large because the profits of being ahead of the queue of limit orders at a given price is larger in this case (in line with the analysis in Chapter 6).

Kirilenko et al. (2017) study the role played by high-frequency traders in the 2010 U.S. equity Flash crash. They do not find that high-frequency traders significantly changed their behavior during the day of the crash relative to other days. As on other days, high-frequency market orders appear to be informed on very short-term price movements. Brogaard et al. (2018) study the behavior of high-frequency traders during stock-specific extreme price movements. They do not find evidence that these traders amplify these movements. Rather, they trade against the direction of these movements.

9.8 Exercises

1. **Private vs. social benefits.** Consider the model of Section 9.4. Suppose that the parameters of the model are as follows: $\pi = 0.3$, $\delta = 2$, $\sigma = 2$,

and $N=5$. Moreover, suppose that the cost of being fast is $C = 0.02$ (one percent of the largest possible profit per trade, σ).

- a. Suppose that all speculators are slow. What is the equilibrium bid-ask spread?
 - b. Show that if the dealer expects all speculators to be slow, at least one speculator is better off paying C to become fast.
 - c. Suppose that all speculators are fast. What is the equilibrium bid-ask spread?
 - d. Show that if the dealer expects all speculators to be fast, all speculators are better off paying C to be fast. Is this situation a Nash equilibrium?
 - e. Show that speculators' expected profit is larger if they are all slow than if they are all fast. Why is this not an equilibrium? What is the problem?
2. **Risk of being picked off and investors' attention.** In limit order markets, all investors, including retail investors, can submit market or limit orders. Benamar (2019) analyzes the performance of retail investors submitting limit orders at a French on-line broker over the period 2002–10. His dataset contains more than 100,000 retail investors over the whole period, 60% of their orders are market orders and 20% of their orders are limit orders (the remaining 20% include more exotic types of orders such as stop-loss orders). In June 2003, the on-line broker began offering a new technology called “TradeRobot,” a computerized system enabling retail investors to better monitor their limit orders (e.g., by obtaining customized alerts conditional on stock movements and sending multiple orders very quickly in reaction to these alerts). Only a subset of retail investors (about 2,000) have used TradeRobot. Let us call “treated group” the group of retail investors using TradeRobot and “control group,” a group of retail investors that never used TradeRobot but are otherwise (in terms of education, age, location trading behavior, etc.) very similar to those in the treatment group.

Each time the limit order (sell or buy) of a retail investor executes, Benamar (2019) computes the following measure:

$$R = d \times \frac{P_c - P_{exe}}{P_{exe}},$$

where (i) P_c is the closing price on the day in which the order is executed, (ii) P_{exe} is the execution price and (iii) $d = +1$ if the limit order is a buy limit order and $d = -1$ if this is a sell order. Figure 9.6 shows the evolution of the **average** monthly value of this measure for investors in the treatment group (thick line) and in the control group (dashed line) from 2001 to 2010 (the light gray line is the value of R for investors who are in none of the two

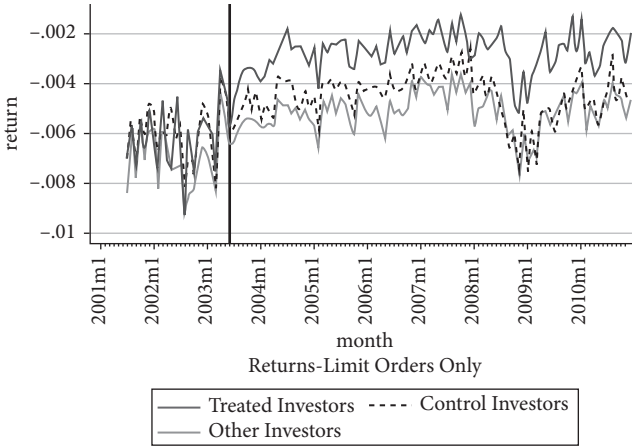


Figure 9.6 Returns to investors from placing limit orders.

Source: Benamar (2019), fig. 4.

previous groups: investors who never used TradeRobot but had different characteristics from investors in the treated group).²⁷

a. How do you explain that the average value of R is always negative for both groups? Is this surprising?

b How do you explain that the average value of R becomes more negative for both groups after 2008?

c. How do you explain that after the introduction of TradeRobot in June 2003 (corresponding to the vertical line in Figure 9.6), the average value of R is smaller in absolute value (though still negative) for investors in the treatment group than for investors in the control group? (Note that the difference between the average value of R for the two groups is statistically significant after June 2003 but not before.)

d. If you were a broker, would the evidence in Figure 9.6 lead you to advise your retail customers to use market orders rather than limit orders? What further information would you need to be able to advise your retail customers about this choice?

²⁷ For instance, suppose that in January 2009, there are two investors, A and B, in the treated group and that for investor A, $R = 2\%$ whereas for investor B, $R = -3\%$. Then, the average value of R for investors in the treated group in January 2009 would be $\frac{1}{2} \times 2\% - \frac{1}{2} \times 3\% = -0.5\%$.